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MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

VII SEMESTER B.TECH. (ELECTRICAL & ELECTRONICS ENGINEERING)
MAKEUP EXAMINATIONS, DECEMBER 2024

DATABASE MANAGEMENT SYSTEMS [ELE 4079]

REVISED CREDIT SYSTEM

Time: 3 Hours **28 December 2024** **Max. Marks: 50**

Instructions to Candidates:

- Answer **ALL** the questions.
- Missing data may be suitably assumed.

1A. Give names of **any THREE** popular database management systems used in real-world applications. ***(03)***

1B. What is a B+ tree? Illustrate the construction of B+ tree of order 3 for data points 1, 2, 3 and 4. Numbers are inserted into the tree in the same sequence. ***(03)***

1C. List **any FOUR** tasks that come under Data Definition Language. ***(04)***

2A. With the help of a suitable example, explain the terms **ENTITY**, **ATTRIBUTE**, **RELATION** as referred to **ENTITY-RELATIONSHIP MODEL (ER)**. ***(03)***

2B. Develop an **ENTITY-RELATIONSHIP (ER)** model with proper notations, for the following requirement:
"A student is enrolled for course taught by a teacher." ***(03)***

2C. For the Entity-Relationship (ER) model developed in **Q2B** give the **RELATIONAL SCHEMA** with proper notations.
Requirement: "A student is enrolled for course taught by a teacher." ***(04)***

3A. Identify the appropriate **RELATIONAL ALGEBRA** operation form the statements given below.

1. Combines every row of one relation with every row of another relation.
2. Selects specific columns (attributes) from a relation.
3. Filters rows from a relation based on a condition. ***(03)***

3B. Consider the following information in table below:

Student_ID	Student_Name	Student_Marks
101	Alice	{Math:90, Science:85}
102	Bob	{Math:88, History:80}

Convert the student information into **1NF**, **2NF** and **3NF**, by applying principles of relational database design. ***(03)***

3C. What is **LOSSLESS DECOMPOSITION**, explain with the help of suitable example. What is the rule for **LOSSLESS DECOMPOSITION**?

(04)

4A. Identify if there are any errors in the following **CREATE TABLE SQL** statement. If errors exist, explain them, and rewrite the statement correctly.

```
CREATE TABLE Employee (
    EmployeeID INT PRIMARY KEY AUTO_INCREMENT,
    Name VARCHAR(50),
    Salary DOUBLE DEFAULT "None",
    DepartmentID INT REFERENCES Departments(DeptID),
    StartDate DATE DEFAULT SYSDATE,
    PRIMARY KEY (EmployeeID)
);
```

(03)

4B. Identify if there are any errors in the following **INSERT TABLE SQL** statement. If errors exist, explain them, and rewrite the statement correctly.

```
INSERT INTO Employee (EmployeeID, Name, Salary, DepartmentID, StartDate) VALUES (1, 'Alice', "fifty thousand", 'HR', 2023-12-01);
```

(03)

4C. Correct the following SELECT SQL statement:

```
SELECT DepartmentID, Name, AVG(Salary)
FROM Employee
GROUP BY Salary
HAVING AVG(Salary) > 50000;
```

Explain how the errors were corrected. ***(04)***

5A. What is a nested subquery in SQL? Rewrite the following query as nested query.

```
SELECT e.name FROM employees e JOIN departments d ON e.department_id = d.department_id WHERE d.department_name = 'Sales';
```

(03)

5B. What is a VIEW in database management systems. Illustrate with suitable SQL statements, how a VIEW is created. Show the usage of the VIEW created using a suitable example. ***(03)***

5C. List any FOUR real-world applications that use database management systems. ***(04)***